Group Member Roles and Responsibilities

Generative Artificial Intelligence — Final Project

Korea University

GLOW vs FLUX: Comparing Normalizing Flows and Flow Matching for Face Image Generation at 64×64 Resolution

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Team composition. This final project was carried out as an **individual (one-person) project**. All roles and responsibilities listed below were performed by the single member.

Member	Affiliation / Contact
Younghyeok Kim	Korea University • `younghyeok25@gmail.com`

Roles and Responsibilities

Role	Responsibilities and Deliverables
Research Design & Project Lead	Defined the research question (normalizing flows vs. flow matching at low resolution), designed the experimental protocol across four evaluation axes (sample quality, exact likelihood, reconstruction fidelity, sampling efficiency), and managed the overall project schedule.
Model Implementation	Implemented GLOW from scratch in PyTorch ($\sim$76M parameters): ActNorm, invertible 1×1 convolution with LU decomposition, affine coupling, and the multi-scale block architecture, plus the FFHQ-64 data pipeline (`model.py`, `dataset.py`).
Training & Infrastructure	Built the training loop (NLL objective, 1,000-step LR warmup, batch 128, 30,000 iterations) and resolved divergence via two stability fixes: log-scale clamping to $[-3, 3]$ and a NaN/Inf gradient guard. Set up checkpointing and a watchdog script for auto-restart on crash (`train.py`, `run_train_with_watchdog.sh`).
Baseline Sampling & Evaluation	Ran pretrained baselines at 64×64 — FLUX.1-schnell/dev (4–28 NFE) and DDIM-10/20/50 / DDPM-100 — and measured FID (clean-fid, 5k samples vs. 70k FFHQ references), exact NLL, reconstruction PSNR/SSIM, and wall-clock sampling cost (`sample_flux.py`, `sample_ddim.py`, `evaluate.py`, `reconstruct.py`).
Analysis & Visualization	Conducted latent-space interpolation, OOD detection via NLL, and 2D two-moons transport-path comparison of the three paradigms; produced all paper figures 1–9 (`interpolation.py`, `ood_detection.py`, `visualize_2d.py`, `make_figures.py`).
Writing & Presentation	Wrote the final report (`report.pdf`, two-column paper format), prepared the presentation slides and script, and authored the code documentation and reproduction guide accompanying the submission.

All implementation, experiments, analysis, and writing were performed solely by Younghyeok Kim.